

Multi-Criteria Decision Analysis Report

Authors

- Mateusz Stawicki 155900
- Mateusz Idziejczak 155842

Car Selection Problem Analysis

1. Dataset Description

We analyzed a set of 50 small city cars available for sale within 100 km of Kozięglowy, with prices under 40,000 PLN. The data comes from [OTOMOTO](#), where we scraped listings for cars that might suit a PUT student's needs. We used whole dataset for this report.

Our *totally fictional* buyer is a student living in Kozięglowy, Wielkopolskie - who needs a reliable city car that balances cost, age, mileage, and performance. Since they're a student, practical considerations outweigh aesthetics or comfort - they want the best value for money and maybe a bit of engine power to impress friends.

Specifically, the dataset comprises the first 50 listings from [this exact url](#)

Not all available scraped data was used for the MCDA in the end. Still, it is present in the dataset for reference.

Full dataset

title	subtitle	price	currency	year	mileage_km	fuel_type	gearbox	location	seller_type	price_evaluation	url
Renault Clio 0.9 Energy TCe Limited	898 cm3 - 90 KM - Salon Polska, Pierwszy właściciel, bezwypadkowy i bezkolizyjny	39500	PLN	2017	60098	Benzyna	Manualna	Rawicz (Wielkopolskie)	Prywatny sprzedawca	Powyżej średniej	https://clio-sa-bezwył
Dacia Sandero Stepway TCe 90 (S&S) Essential	898 cm3 - 90 KM	39360	PLN	2019	65900	Benzyna	Manualna	Dobrcz (Kujawsko-pomorskie)	Prywatny sprzedawca	W granicach średniej	https://sander
...											

MCDA dataset

title	price ↓ (4,700-39,500 PLN)	year ↑ (2000-2019)	mileage_km ↓ (49,459-374,000 km)	engine_size_cm3 ↑ (875-1,798 cm³)	power_hp ↑ (60-160 HP)
Renault Clio 0.9 Energy TCe Limited	39500	2017	60098	898	90
Renault Megane 1.4 RN 16V	4700	2001	173117	1390	95
Dacia Sandero Stepway TCe 90 (S&S) Essential	39360	2019	65900	898	90
Mitsubishi Space Star	26000	2017	82000	1193	80
Nissan Micra	16500	2011	177491	1198	80
Ford KA	19300	2014	139990	1242	69
Alfa Romeo Mito 0.9 TwinAir Progression	7800	2009	232000	875	85
Fiat 500 1.2 8V Anniversario	13900	2012	93000	1242	69
Kia Picanto	36900	2018	56213	998	67
Audi A3	22500	2000	374000	1798	160

2. UTA Method Implementation and Results

2.1 Implementation Overview

For our car selection problem, we implemented the UTA method to analyze the multiple criteria involved. Our implementation includes:

- Using PuLP with the GLPK solver to formulate and solve the linear programming problem
- Developing a mechanism to handle inconsistent preference information
- Adding constraints to prevent any criterion from having a weight greater than 0.5
- Creating artificial alternatives that vary only on pairs of criteria to ensure balanced weights

- Implementing a discriminant value function that maximizes the distance between utilities of alternatives in preference relationships

2.2 Preference Structure

Looking at the cars, the Audi A3 stands out as particularly problematic. Despite having the largest engine and most power, it's from 2000 with a staggering 374,000 km on the odometer - more than double most other options. The Kia Picanto (2018, 56,213 km) and Fiat 500 (2012, 93,000 km) are much newer with far less wear.

The Mitsubishi Space Star seems to be the best overall option:

- Price: 26,000 PLN
- Year: 2017
- Mileage: 82,000 km
- Engine: 1,193 cm³
- Power: 80 HP

It hits a sweet spot with reasonable price, recent year, manageable mileage, and decent performance - perfect for a student who needs reliability without breaking the bank.

We defined the following preference relations to guide our model:

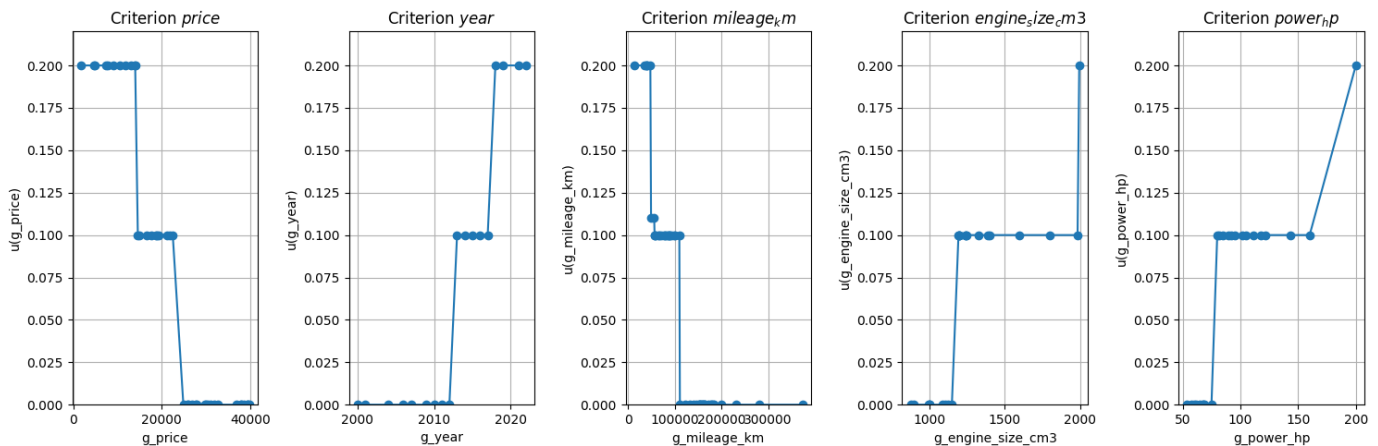
- Alternative_8 > Alternative_9 (Kia Picanto > Audi A3)
- Alternative_7 > Alternative_9 (Fiat 500 > Audi A3)
- Alternative_31 > Alternative_9 (Mitsubishi Space Star > Audi A3)
- Alternative_31 > Alternative_8 (Mitsubishi Space Star > Kia Picanto)
- Alternative_31 > Alternative_7 (Mitsubishi Space Star > Fiat 500)

We also introduced an inconsistent preference:

- Alternative_9 > Alternative_31 (Audi A3 > Mitsubishi Space Star)

This creates a cycle (Mitsubishi > Kia Picanto > Audi A3 > Mitsubishi), which our solver had to find. And it found and removed the contradictory preference (Audi A3 > Mitsubishi Space Star) to fix the cycle in our preference statements.

2.3 Results Analysis



Objective Function

We implemented an objective function that maximizes the sum of utility differences between alternatives in preference relationships:

Maximize: $\sum (U(a) - U(b))$ for all preference relations $a > b$

This approach finds the most discriminant value function by ensuring that preferred alternatives have significantly higher utility values than less preferred ones. Our objective function value was 1.5000, showing that our model successfully maximized the discrimination between alternatives in preference relationships:

1. **Clear Utility Separation:** Looking at our consistent preference relations:

- Alternative_8 > Alternative_9: Utilities 0.31 > 0.30 (difference: 0.01)
- Alternative_7 > Alternative_9: Utilities 0.40 > 0.30 (difference: 0.10)
- Alternative_31 > Alternative_9: Utilities 0.80 > 0.30 (difference: 0.50)
- Alternative_31 > Alternative_8: Utilities 0.80 > 0.31 (difference: 0.49)
- Alternative_31 > Alternative_7: Utilities 0.80 > 0.40 (difference: 0.40)

2. **Total Discrimination:** The sum of these differences (our objective function value) is 1.50, which represents how well our model discriminates between alternatives in preference relationships.

3. **Distinct Utility Clusters:** The utility values show clear groupings (0.80, 0.60, 0.50, 0.40, 0.30-0.31, 0.20, 0.10), indicating that the model separates alternatives into distinct tiers.

Criteria Weights

Our model gave equal weight to all criteria:

- price: 0.2000
- year: 0.2000
- mileage_km: 0.2000
- engine_size_cm3: 0.2000
- power_hp: 0.2000

This happened because of our artificial alternatives that prevent any single criterion from becoming too important. Based on our preference statements, the model couldn't find a reason to make one criterion more important than others.

Car Rankings

Our model successfully found a solution after removing the inconsistent preference, showing that our approach was reasonable. The Mitsubishi Space Star stands out as the clear winner with a utility score of 0.80, which is significantly higher than the other cars.

Top 10 cars according to our model:

1. Mitsubishi Space Star 1.2 Clear Tec CVT Active+ (0.80)
2. Peugeot 208 PureTech 75 Allure (0.60)
3. BMW Seria 1 118i (0.50)
4. Citroën C3 1.2 VTi Attraction (0.50)
5. Fiat 500 1.2 8V Start&Stopp Collezione (0.50)
6. Peugeot 208 1.2 PureTech Style (0.40)
7. Audi A1 1.4 TFSI Sportback S tronic Ambition (0.40)
8. Volkswagen Polo 1.2 TSI BMT Highline (0.40)
9. Suzuki Swift 1.0 T Elegance (0.40)
10. Kia Rio (0.40)

The scores range from 0.10 to 0.80, giving us a good way to tell the cars apart.

The UTA method was effective in analyzing this complex car selection problem by handling the preference inconsistency and giving us a clear ranking of all alternatives.

3. Comparison Between UTA and PROMETHEE Methods

In our previous project, we analyzed the same car selection problem using the PROMETHEE method. Here's how the results compare:

3.1 Results Comparison

Both methods identified the Mitsubishi Space Star as the top choice:

Car Model	UTA Rank	PROMETHEE Rank
Mitsubishi Space Star	1	1
Fiat 500	5	4
Dacia Sandero	6	2
Renault Clio	8	3
Kia Picanto	10	5
Audi A3	16	10

3.2 Key Insights

1. **Consistency at Extremes:** Both methods ranked Mitsubishi Space Star at the top and Audi A3 at the bottom, confirming these as the best and worst options respectively.
2. **Different Middle Rankings:** The methods differed in middle alternatives with Dacia Sandero ranked 6th in UTA but 2nd in PROMETHEE, and Renault Clio ranked 8th in UTA but 3rd in PROMETHEE.
3. **Weight Determination:** UTA determined equal weights (0.2) for all criteria based on our preference statements, while PROMETHEE used our manually assigned weights (Mileage: 3, Price/Year: 2, Engine/Power: 1).
4. **Confidence in Top Choice:** Both methods converging on the Mitsubishi Space Star as the best option gives us high confidence in this recommendation for our student car buyer.

4. AHP Method Implementation and Results

4.1 Implementation Overview

For our car selection problem, we also implemented the Analytic Hierarchy Process (AHP) method to provide another perspective. Our implementation includes:

- Creating a three-level hierarchy with the goal (car selection) at the top level

- Defining three main categories at the second level: Cost, Performance, and Reliability
- Placing five criteria at the third level, distributed across the categories:
 - Cost: Price
 - Performance: Engine Size, Power
 - Reliability: Year, Mileage
- Using the eigenvector method to calculate weights and consistency ratios
- Implementing consistency checks for all pairwise comparison matrices
- Analyzing inconsistencies by reconstructing matrices from calculated weights

4.2 Pairwise Comparison Matrices

We created pairwise comparison matrices for each level of the hierarchy using the standard 1-9 scale:

Level 1: Categories

	Reliability	Cost	Performance
Reliability	1.000000	3.000000	5.000000
Cost	0.333333	1.000000	3.000000
Performance	0.250000	0.333333	1.000000

This matrix was intentionally made inconsistent (Performance vs. Reliability should be 1/5, but we used 1/4). The consistency ratio was 0.1174, which exceeds the acceptable threshold of 0.1.

Level 2: Performance Criteria

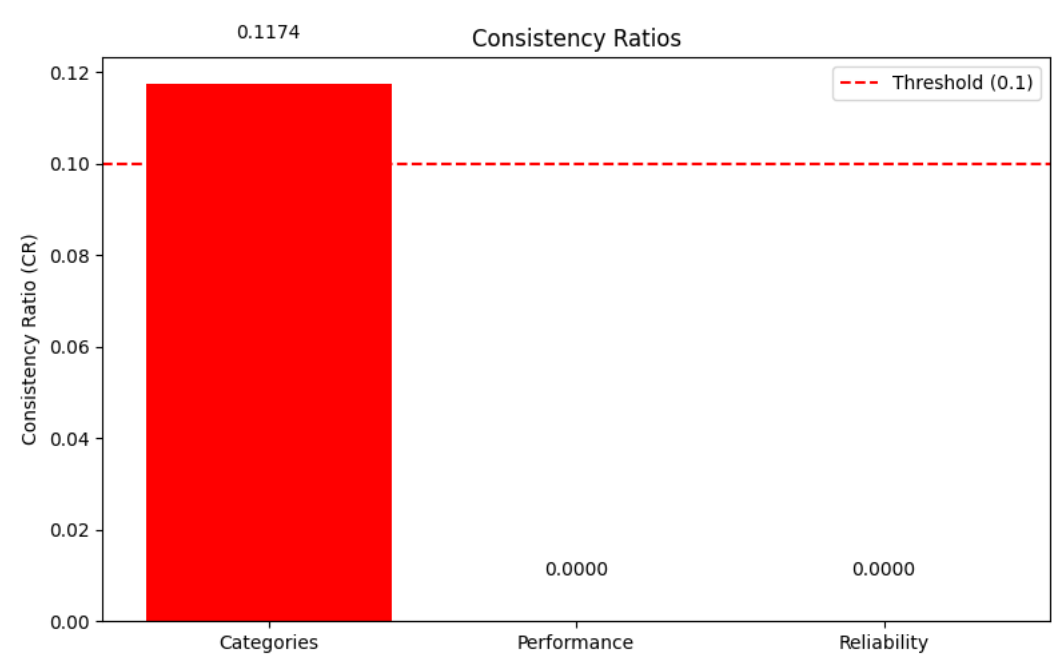
	Engine Size	Power
Engine Size	1.000000	0.333333
Power	3.000000	1.000000

Level 2: Reliability Criteria

	Mileage	Year
Mileage	1.000000	3.000000
Year	0.333333	1.000000

Both Level 2 matrices were perfectly consistent with a consistency ratio of 0.0000.

4.3 Inconsistency Analysis



For the inconsistent Level 1 matrix, we reconstructed it from the calculated weights:

	Reliability	Cost	Performance
Reliability	1.000000	2.437196	5.523185
Cost	0.410308	1.000000	2.266205
Performance	0.181055	0.441266	1.000000

The difference matrix (Original - Reconstructed) revealed:

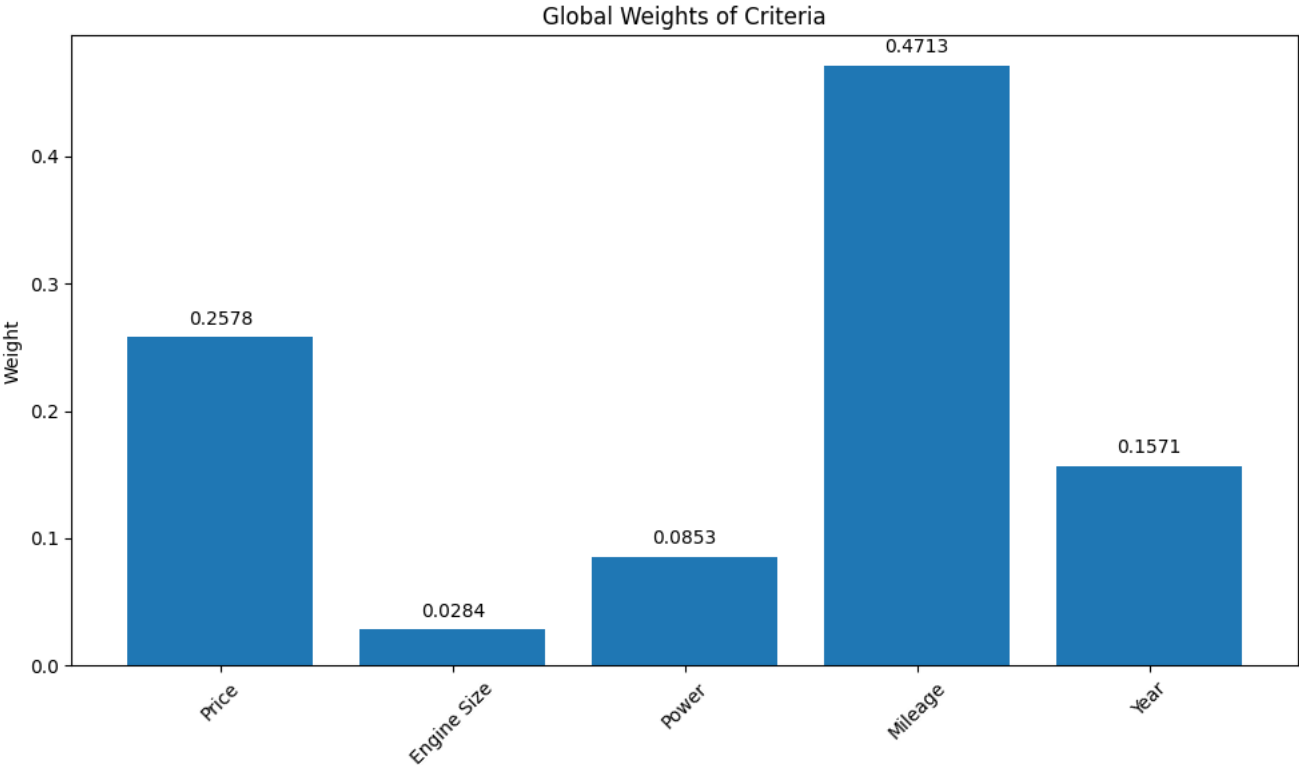
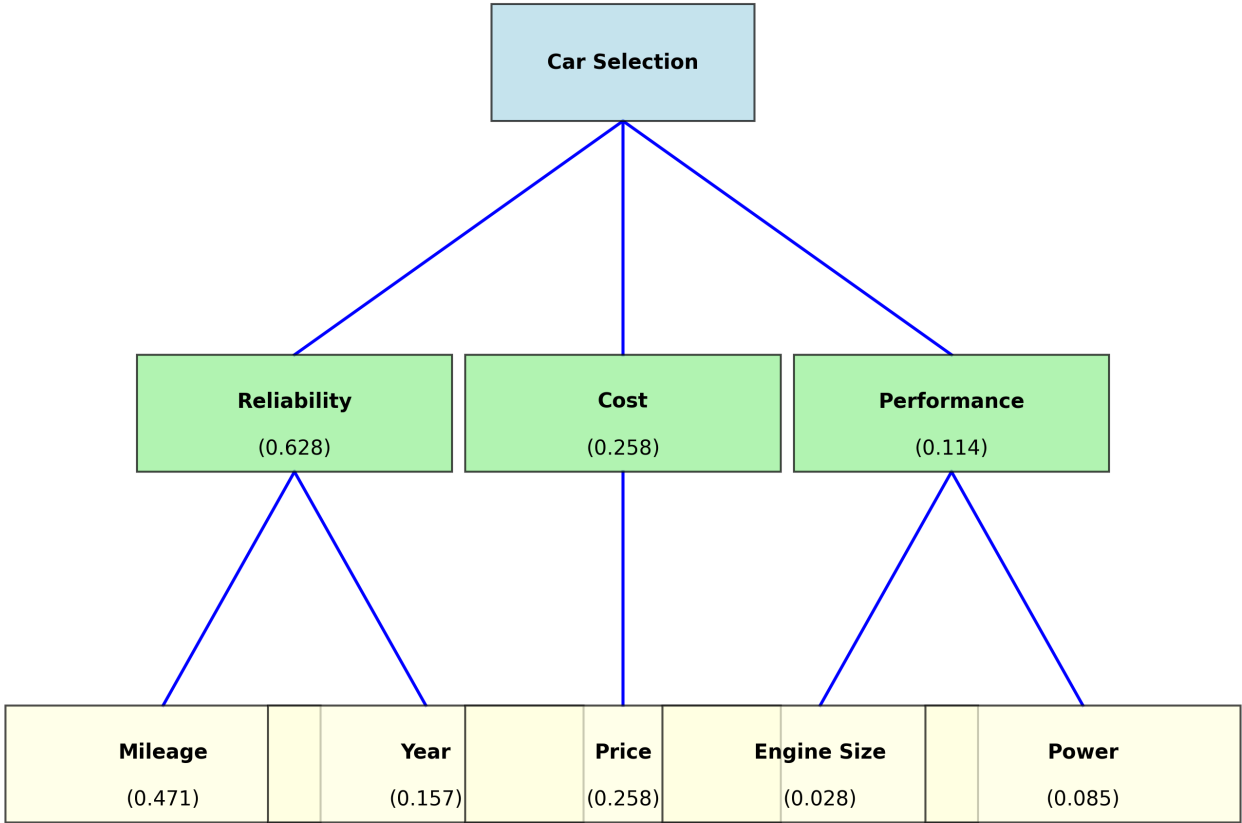
	Reliability	Cost	Performance
Reliability	0.000000	0.562804	-0.523185
Cost	-0.076974	0.000000	0.733795
Performance	0.068945	-0.107933	0.000000

The largest inconsistency was 0.7338 at the position "Cost vs. Performance", confirming that our intentional inconsistency was detected.

4.4 Weights and Results

The AHP method produced the following weights for our criteria:

Category	Local Weight	Criterion	Local Weight	Global Weight
Reliability	0.6284	Mileage	0.7500	0.4713
Reliability	0.6284	Year	0.2500	0.1571
Cost	0.2578	Price	1.0000	0.2578
Performance	0.1138	Power	0.7500	0.0853
Performance	0.1138	Engine Size	0.2500	0.0284



These weights show that Mileage was considered the most important criterion (47.13%), followed by Price (25.78%) and Year (15.71%), with Power (8.53%) and Engine Size (2.84%) being less important.

The top 10 cars according to AHP were:

- 1. Mitsubishi Space Star 1.2 Clear Tec CVT Active+ (0.786)
- 2. Kia Picanto (0.782)
- 3. Peugeot 208 PureTech 75 Allure (0.777)
- 4. Nissan Micra 1.2 City (0.765)
- 5. Peugeot 206 plus 206+ 60 Generation (0.761)

- #### 4.5 Comparison with UTA and PROMETHEE Results

- Kia Picanto ranked 2nd in AHP, but 10th in UTA and 5th in PROMETHEE
- Peugeot 208 ranked 3rd in AHP, but 2nd in UTA
- Dacia Sandero ranked 38th in AHP, but 6th in UTA and 2nd in PROMETHEE

- AHP heavily weighted Mileage (47.13%) and Price (25.78%)
- UTA assigned equal weights (20%) to all criteria
- PROMETHEE used manually assigned weights with emphasis on Mileage

Appendix A

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Preferences:  
Alternative_8 > Alternative_9  
Alternative_7 > Alternative_9  
Alternative_31 > Alternative_9  
Alternative_31 > Alternative_8  
Alternative_31 > Alternative_7  
Alternative_9 > Alternative_31  
Found consistent subset with 5 preferences  
Found consistent subset with 4 preferences  
Found consistent subset with 4 preferences  
Found consistent subset with 4 preferences  
Found consistent subset with 4 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 3 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
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Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 2 preferences  
Found consistent subset with 1 preferences  
Found consistent subset with 1 preferences  
Found consistent subset with 1 preferences  
Found consistent subset with 1 preferences  
Found consistent subset with 1 preferences
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Alternative_31 > Alternative_8
Alternative_31 > Alternative_7

Inconsistent preferences:
Alternative_9 > Alternative_31

Solution status: Optimal

Objective function value: 1.5000

Objective Function:
  3*Alternative_31_Utility + 0*Alternative_7_Utility + 0*Alternative_8_Utility - 3*Alternative_9_Utility

All Constraints:
Monotonicity_price_0: price_1700 - price_4700 >= 0
Monotonicity_price_1: price_4700 - price_4900 >= 0
Monotonicity_price_2: price_4900 - price_7400 >= 0
Monotonicity_price_3: price_7400 - price_7800 >= 0
Monotonicity_price_4: price_7800 - price_9000 >= 0
Monotonicity_price_5: -price_10500 + price_9000 >= 0
Monotonicity_price_6: price_10500 - price_11800 >= 0
Monotonicity_price_7: price_11800 - price_12999 >= 0
Monotonicity_price_8: price_12999 - price_13900 >= 0
Monotonicity_price_9: price_13900 - price_13999 >= 0
Monotonicity_price_10: price_13999 - price_14600 >= 0
Monotonicity_price_11: price_14600 - price_15000 >= 0
Monotonicity_price_12: price_15000 - price_16500 >= 0
Monotonicity_price_13: price_16500 - price_17700 >= 0
Monotonicity_price_14: price_17700 - price_18700 >= 0
Monotonicity_price_15: price_18700 - price_19000 >= 0
Monotonicity_price_16: price_19000 - price_19300 >= 0
Monotonicity_price_17: price_19300 - price_21300 >= 0
Monotonicity_price_18: price_21300 - price_21900 >= 0
Monotonicity_price_19: price_21900 - price_22500 >= 0
Monotonicity_price_20: price_22500 - price_24900 >= 0
Monotonicity_price_21: price_24900 - price_25900 >= 0
Monotonicity_price_22: price_25900 - price_26000 >= 0
Monotonicity_price_23: price_26000 - price_26900 >= 0
Monotonicity_price_24: price_26900 - price_27700 >= 0
Monotonicity_price_25: price_27700 - price_27999 >= 0
Monotonicity_price_26: price_27999 - price_29800 >= 0
Monotonicity_price_27: price_29800 - price_29900 >= 0
Monotonicity_price_28: price_29900 - price_30500 >= 0
Monotonicity_price_29: price_30500 - price_31000 >= 0
Monotonicity_price_30: price_31000 - price_31900 >= 0
Monotonicity_price_31: price_31900 - price_32700 >= 0
Monotonicity_price_32: price_32700 - price_36900 >= 0
Monotonicity_price_33: price_36900 - price_37900 >= 0
Monotonicity_price_34: price_37900 - price_38000 >= 0
Monotonicity_price_35: price_38000 - price_38500 >= 0
Monotonicity_price_36: price_38500 - price_39360 >= 0
Monotonicity_price_37: price_39360 - price_39500 >= 0
Monotonicity_price_38: price_39500 - price_39850 >= 0
Monotonicity_year_0: year_2000 - year_2001 <= 0
Monotonicity_year_1: year_2001 - year_2004 <= 0
Monotonicity_year_2: year_2004 - year_2006 <= 0
Monotonicity_year_3: year_2006 - year_2007 <= 0
Monotonicity_year_4: year_2007 - year_2009 <= 0
Monotonicity_year_5: year_2009 - year_2010 <= 0
Monotonicity_year_6: year_2010 - year_2011 <= 0
Monotonicity_year_7: year_2011 - year_2012 <= 0
Monotonicity_year_8: year_2012 - year_2013 <= 0
Monotonicity_year_9: year_2013 - year_2014 <= 0
Monotonicity_year_10: year_2014 - year_2015 <= 0
Monotonicity_year_11: year_2015 - year_2016 <= 0
Monotonicity_year_12: year_2016 - year_2017 <= 0
Monotonicity_year_13: year_2017 - year_2018 <= 0
Monotonicity_year_14: year_2018 - year_2019 <= 0
Monotonicity_year_15: year_2019 - year_2021 <= 0
Monotonicity_year_16: year_2021 - year_2022 <= 0
Monotonicity_mileage_km_0: mileage_km_13000 - mileage_km_34900 >= 0
Monotonicity_mileage_km_1: mileage_km_34900 - mileage_km_39000 >= 0
Monotonicity_mileage_km_2: mileage_km_39000 - mileage_km_40400 >= 0
Monotonicity_mileage_km_3: mileage_km_40400 - mileage_km_42000 >= 0
Monotonicity_mileage_km_4: mileage_km_42000 - mileage_km_48000 >= 0
Monotonicity_mileage_km_5: mileage_km_48000 - mileage_km_49459 >= 0
Monotonicity_mileage_km_6: mileage_km_49459 - mileage_km_56213 >= 0
Monotonicity_mileage_km_7: mileage_km_56213 - mileage_km_57000 >= 0
Monotonicity_mileage_km_8: mileage_km_57000 - mileage_km_58000 >= 0
Monotonicity_mileage_km_9: mileage_km_58000 - mileage_km_60098 >= 0
Monotonicity_mileage_km_10: mileage_km_60098 - mileage_km_65900 >= 0

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Monotonicity_mileage_km_11: mileage_km_65900 - mileage_km_68000 >= 0
Monotonicity_mileage_km_12: mileage_km_68000 - mileage_km_71662 >= 0
Monotonicity_mileage_km_13: mileage_km_71662 - mileage_km_78000 >= 0
Monotonicity_mileage_km_14: mileage_km_78000 - mileage_km_79000 >= 0
Monotonicity_mileage_km_15: mileage_km_79000 - mileage_km_82000 >= 0
Monotonicity_mileage_km_16: mileage_km_82000 - mileage_km_85000 >= 0
Monotonicity_mileage_km_17: mileage_km_85000 - mileage_km_86000 >= 0
Monotonicity_mileage_km_18: mileage_km_86000 - mileage_km_87975 >= 0
Monotonicity_mileage_km_19: mileage_km_87975 - mileage_km_88000 >= 0
Monotonicity_mileage_km_20: mileage_km_88000 - mileage_km_93000 >= 0
Monotonicity_mileage_km_21: mileage_km_93000 - mileage_km_99600 >= 0
Monotonicity_mileage_km_22: -mileage_km_101000 + mileage_km_99600 >= 0
Monotonicity_mileage_km_23: mileage_km_101000 - mileage_km_110000 >= 0
Monotonicity_mileage_km_24: mileage_km_110000 - mileage_km_111200 >= 0
Monotonicity_mileage_km_25: mileage_km_111200 - mileage_km_122185 >= 0
Monotonicity_mileage_km_26: mileage_km_122185 - mileage_km_132000 >= 0
Monotonicity_mileage_km_27: mileage_km_132000 - mileage_km_139990 >= 0
Monotonicity_mileage_km_28: mileage_km_139990 - mileage_km_145500 >= 0
Monotonicity_mileage_km_29: mileage_km_145500 - mileage_km_152000 >= 0
Monotonicity_mileage_km_30: mileage_km_152000 - mileage_km_153000 >= 0
Monotonicity_mileage_km_31: mileage_km_153000 - mileage_km_153533 >= 0
Monotonicity_mileage_km_32: mileage_km_153533 - mileage_km_154400 >= 0
Monotonicity_mileage_km_33: mileage_km_154400 - mileage_km_158000 >= 0
Monotonicity_mileage_km_34: mileage_km_158000 - mileage_km_159000 >= 0
Monotonicity_mileage_km_35: mileage_km_159000 - mileage_km_160000 >= 0
Monotonicity_mileage_km_36: mileage_km_160000 - mileage_km_163035 >= 0
Monotonicity_mileage_km_37: mileage_km_163035 - mileage_km_164000 >= 0
Monotonicity_mileage_km_38: mileage_km_164000 - mileage_km_165000 >= 0
Monotonicity_mileage_km_39: mileage_km_165000 - mileage_km_173117 >= 0
Monotonicity_mileage_km_40: mileage_km_173117 - mileage_km_177491 >= 0
Monotonicity_mileage_km_41: mileage_km_177491 - mileage_km_177896 >= 0
Monotonicity_mileage_km_42: mileage_km_177896 - mileage_km_181500 >= 0
Monotonicity_mileage_km_43: mileage_km_181500 - mileage_km_184000 >= 0
Monotonicity_mileage_km_44: mileage_km_184000 - mileage_km_199700 >= 0
Monotonicity_mileage_km_45: mileage_km_199700 - mileage_km_232000 >= 0
Monotonicity_mileage_km_46: mileage_km_232000 - mileage_km_281980 >= 0
Monotonicity_mileage_km_47: mileage_km_281980 - mileage_km_374000 >= 0
Monotonicity_engine_size_cm3_0: engine_size_cm3_875 - engine_size_cm3_898 <= 0
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Monotonicity_engine_size_cm3_5: engine_size_cm3_1084 - engine_size_cm3_1108 <= 0
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Monotonicity_engine_size_cm3_10: engine_size_cm3_1197 - engine_size_cm3_1198 <= 0
Monotonicity_engine_size_cm3_11: engine_size_cm3_1198 - engine_size_cm3_1199 <= 0
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Monotonicity_engine_size_cm3_14: engine_size_cm3_1242 - engine_size_cm3_1248 <= 0
Monotonicity_engine_size_cm3_15: engine_size_cm3_1248 - engine_size_cm3_1328 <= 0
Monotonicity_engine_size_cm3_16: engine_size_cm3_1328 - engine_size_cm3_1390 <= 0
Monotonicity_engine_size_cm3_17: engine_size_cm3_1390 - engine_size_cm3_1398 <= 0
Monotonicity_engine_size_cm3_18: engine_size_cm3_1398 - engine_size_cm3_1595 <= 0
Monotonicity_engine_size_cm3_19: engine_size_cm3_1595 - engine_size_cm3_1798 <= 0
Monotonicity_engine_size_cm3_20: engine_size_cm3_1798 - engine_size_cm3_1984 <= 0
Monotonicity_engine_size_cm3_21: engine_size_cm3_1984 - engine_size_cm3_1995 <= 0
Monotonicity_power_hp_0: power_hp_54 - power_hp_58 <= 0
Monotonicity_power_hp_1: power_hp_58 - power_hp_60 <= 0
Monotonicity_power_hp_2: power_hp_60 - power_hp_62 <= 0
Monotonicity_power_hp_3: power_hp_62 - power_hp_65 <= 0
Monotonicity_power_hp_4: power_hp_65 - power_hp_67 <= 0
Monotonicity_power_hp_5: power_hp_67 - power_hp_68 <= 0
Monotonicity_power_hp_6: power_hp_68 - power_hp_69 <= 0
Monotonicity_power_hp_7: power_hp_69 - power_hp_75 <= 0
Monotonicity_power_hp_8: power_hp_75 - power_hp_80 <= 0
Monotonicity_power_hp_9: power_hp_80 - power_hp_82 <= 0
Monotonicity_power_hp_10: power_hp_82 - power_hp_85 <= 0
Monotonicity_power_hp_11: power_hp_85 - power_hp_90 <= 0
Monotonicity_power_hp_12: power_hp_90 - power_hp_92 <= 0
Monotonicity_power_hp_13: power_hp_92 - power_hp_95 <= 0
Monotonicity_power_hp_14: -power_hp_102 + power_hp_95 <= 0
Monotonicity_power_hp_15: power_hp_102 - power_hp_105 <= 0
Monotonicity_power_hp_16: power_hp_105 - power_hp_111 <= 0
Monotonicity_power_hp_17: power_hp_111 - power_hp_118 <= 0
Monotonicity_power_hp_18: power_hp_118 - power_hp_122 <= 0
Monotonicity_power_hp_19: power_hp_122 - power_hp_143 <= 0
Monotonicity_power_hp_20: power_hp_143 - power_hp_160 <= 0
Monotonicity_power_hp_21: power_hp_160 - power_hp_200 <= 0
```

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Normalize_price_Worst: price_39850 = 0
Normalize_year_Worst: year_2000 = 0
Normalize_mileage_km_Worst: mileage_km_374000 = 0
Normalize_engine_size_cm3_Worst: engine_size_cm3_875 = 0
Normalize_power_hp_Worst: power_hp_54 = 0
Normalize_Sum_Best: engine_size_cm3_1995 + mileage_km_13000 + power_hp_200 + price_1700 + year_2022 = 1.0
Weight_price_Max: price_1700 <= 0.5
Weight_price_Min: price_1700 >= 0.1
Weight_year_Max: year_2022 <= 0.5
Weight_year_Min: year_2022 >= 0.1
Weight_mileage_km_Max: mileage_km_13000 <= 0.5
Weight_mileage_km_Min: mileage_km_13000 >= 0.1
Weight_engine_size_cm3_Max: engine_size_cm3_1995 <= 0.5
Weight_engine_size_cm3_Min: engine_size_cm3_1995 >= 0.1
Weight_power_hp_Max: power_hp_200 <= 0.5
Weight_power_hp_Min: power_hp_200 >= 0.1
Utility_Alternative_0: Alternative_0_Utility - engine_size_cm3_898 - mileage_km_60098 - power_hp_90 - price_39500
- year_2017 = -0.0
Utility_Alternative_1: Alternative_1_Utility - engine_size_cm3_1390 - mileage_km_173117 - power_hp_95 - price_4700
- year_2001 = -0.0
Utility_Alternative_2: Alternative_2_Utility - engine_size_cm3_898 - mileage_km_65900 - power_hp_90 - price_39360
- year_2019 = -0.0
Utility_Alternative_3: Alternative_3_Utility - engine_size_cm3_1193 - mileage_km_82000 - power_hp_80 - price_26000
- year_2017 = -0.0
Utility_Alternative_4: Alternative_4_Utility - engine_size_cm3_1198 - mileage_km_177491 - power_hp_80 -
price_16500 - year_2011 = -0.0
Utility_Alternative_5: Alternative_5_Utility - engine_size_cm3_1242 - mileage_km_139990 - power_hp_69 -
price_19300 - year_2014 = -0.0
Utility_Alternative_6: Alternative_6_Utility - engine_size_cm3_875 - mileage_km_232000 - power_hp_85 - price_7800
- year_2009 = -0.0
Utility_Alternative_7: Alternative_7_Utility - engine_size_cm3_1242 - mileage_km_93000 - power_hp_69 - price_13900
- year_2012 = -0.0
Utility_Alternative_8: Alternative_8_Utility - engine_size_cm3_998 - mileage_km_56213 - power_hp_67 - price_36900
- year_2018 = -0.0
Utility_Alternative_9: Alternative_9_Utility - engine_size_cm3_1798 - mileage_km_374000 - power_hp_160 -
price_22500 - year_2000 = -0.0
Utility_Alternative_10: Alternative_10_Utility - engine_size_cm3_898 - mileage_km_71662 - power_hp_90 -
price_32700 - year_2016 = -0.0
Utility_Alternative_11: Alternative_11_Utility - engine_size_cm3_998 - mileage_km_101000 - power_hp_111 -
price_38000 - year_2018 = -0.0
Utility_Alternative_12: Alternative_12_Utility - engine_size_cm3_1197 - mileage_km_154400 - power_hp_105 -
price_25900 - year_2013 = -0.0
Utility_Alternative_13: Alternative_13_Utility - engine_size_cm3_998 - mileage_km_49459 - power_hp_69 -
price_31000 - year_2014 = -0.0
Utility_Alternative_14: Alternative_14_Utility - engine_size_cm3_1595 - mileage_km_281980 - power_hp_102 -
price_14600 - year_2007 = -0.0
Utility_Alternative_15: Alternative_15_Utility - engine_size_cm3_1199 - mileage_km_13000 - power_hp_75 -
price_21900 - year_2021 = -0.0
Utility_Alternative_16: Alternative_16_Utility - engine_size_cm3_1390 - mileage_km_86000 - power_hp_122 -
price_37900 - year_2013 = -0.0
Utility_Alternative_17: Alternative_17_Utility - engine_size_cm3_999 - mileage_km_159000 - power_hp_60 -
price_18700 - year_2014 = -0.0
Utility_Alternative_18: Alternative_18_Utility - engine_size_cm3_875 - mileage_km_88000 - power_hp_85 -
price_29900 - year_2017 = -0.0
Utility_Alternative_19: Alternative_19_Utility - engine_size_cm3_898 - mileage_km_57000 - power_hp_90 -
price_31900 - year_2019 = -0.0
Utility_Alternative_20: Alternative_20_Utility - engine_size_cm3_999 - mileage_km_68000 - power_hp_62 - price_9000
- year_2010 = -0.0
Utility_Alternative_21: Alternative_21_Utility - engine_size_cm3_1149 - mileage_km_132000 - power_hp_75 -
price_13999 - year_2013 = -0.0
Utility_Alternative_22: Alternative_22_Utility - engine_size_cm3_995 - mileage_km_79000 - power_hp_68 -
price_12999 - year_2011 = -0.0
Utility_Alternative_23: Alternative_23_Utility - engine_size_cm3_1197 - mileage_km_122185 - power_hp_118 -
price_39850 - year_2017 = -0.0
Utility_Alternative_24: Alternative_24_Utility - engine_size_cm3_1248 - mileage_km_34900 - power_hp_85 -
price_29800 - year_2012 = -0.0
Utility_Alternative_25: Alternative_25_Utility - engine_size_cm3_1197 - mileage_km_160000 - power_hp_105 -
price_30500 - year_2015 = -0.0
Utility_Alternative_26: Alternative_26_Utility - engine_size_cm3_1124 - mileage_km_177896 - power_hp_75 -
price_11800 - year_2011 = -0.0
Utility_Alternative_27: Alternative_27_Utility - engine_size_cm3_1398 - mileage_km_165000 - power_hp_90 -
price_29900 - year_2017 = -0.0
Utility_Alternative_28: Alternative_28_Utility - engine_size_cm3_1124 - mileage_km_78000 - power_hp_60 -
price_10500 - year_2009 = -0.0
Utility_Alternative_29: Alternative_29_Utility - engine_size_cm3_998 - mileage_km_158000 - power_hp_69 -
price_17700 - year_2015 = -0.0
Utility_Alternative_30: Alternative_30_Utility - engine_size_cm3_1199 - mileage_km_78000 - power_hp_82 -
price_25900 - year_2017 = -0.0
Utility_Alternative_31: Alternative_31_Utility - engine_size_cm3_1193 - mileage_km_48000 - power_hp_80 -
price_13999 - year_2019 = -0.0
Utility_Alternative_32: Alternative_32_Utility - engine_size_cm3_1197 - mileage_km_58000 - power_hp_90 -

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price_31900 - year_2015 = -0.0
Utility_Alternative_33: Alternative_33_Utility - engine_size_cm3_998 - mileage_km_152000 - power_hp_69 -
price_13900 - year_2007 = -0.0
Utility_Alternative_34: Alternative_34_Utility - engine_size_cm3_1398 - mileage_km_181500 - power_hp_75 -
price_22500 - year_2015 = -0.0
Utility_Alternative_35: Alternative_35_Utility - engine_size_cm3_999 - mileage_km_87975 - power_hp_60 -
price_38500 - year_2019 = -0.0
Utility_Alternative_36: Alternative_36_Utility - engine_size_cm3_1199 - mileage_km_39000 - power_hp_82 -
price_37900 - year_2016 = -0.0
Utility_Alternative_37: Alternative_37_Utility - engine_size_cm3_1149 - mileage_km_153000 - power_hp_58 -
price_7400 - year_2010 = -0.0
Utility_Alternative_38: Alternative_38_Utility - engine_size_cm3_1084 - mileage_km_153533 - power_hp_75 -
price_29900 - year_2022 = -0.0
Utility_Alternative_39: Alternative_39_Utility - engine_size_cm3_1242 - mileage_km_42000 - power_hp_69 -
price_29900 - year_2018 = -0.0
Utility_Alternative_40: Alternative_40_Utility - engine_size_cm3_1995 - mileage_km_110000 - power_hp_143 -
price_16500 - year_2007 = -0.0
Utility_Alternative_41: Alternative_41_Utility - engine_size_cm3_999 - mileage_km_40400 - power_hp_60 -
price_27700 - year_2017 = -0.0
Utility_Alternative_42: Alternative_42_Utility - engine_size_cm3_998 - mileage_km_85000 - power_hp_65 -
price_27999 - year_2021 = -0.0
Utility_Alternative_43: Alternative_43_Utility - engine_size_cm3_1984 - mileage_km_164000 - power_hp_200 -
price_26900 - year_2011 = -0.0
Utility_Alternative_44: Alternative_44_Utility - engine_size_cm3_1108 - mileage_km_199700 - power_hp_54 -
price_1700 - year_2006 = -0.0
Utility_Alternative_45: Alternative_45_Utility - engine_size_cm3_1328 - mileage_km_145500 - power_hp_92 -
price_15000 - year_2009 = -0.0
Utility_Alternative_46: Alternative_46_Utility - engine_size_cm3_1199 - mileage_km_184000 - power_hp_82 -
price_21300 - year_2013 = -0.0
Utility_Alternative_47: Alternative_47_Utility - engine_size_cm3_998 - mileage_km_99600 - power_hp_69 -
price_24900 - year_2011 = -0.0
Utility_Alternative_48: Alternative_48_Utility - engine_size_cm3_1240 - mileage_km_111200 - power_hp_80 -
price_4900 - year_2004 = -0.0
Utility_Alternative_49: Alternative_49_Utility - engine_size_cm3_998 - mileage_km_163035 - power_hp_80 -
price_19000 - year_2013 = -0.0
Pref_Alternative_8_Alternative_9: Alternative_8_Utility - Alternative_9_Utility >= 0.01
Pref_Alternative_7_Alternative_9: Alternative_7_Utility - Alternative_9_Utility >= 0.01
Pref_Alternative_31_Alternative_9: Alternative_31_Utility - Alternative_9_Utility >= 0.01
Pref_Alternative_31_Alternative_8: Alternative_31_Utility - Alternative_8_Utility >= 0.01
Pref_Alternative_31_Alternative_7: Alternative_31_Utility - Alternative_7_Utility >= 0.01
ArtificialNotDictatorial_price_year_1: price_1700 - price_39850 + year_2000 - year_2022 <= 0.95
ArtificialNotDictatorial_price_year_2: -price_1700 + price_39850 - year_2000 + year_2022 <= 0.95
ArtificialNotFlat_price_year_1: price_22500 - price_39850 >= 0.1
ArtificialNotFlat_price_year_2: price_1700 - price_22500 >= 0.1
ArtificialNotFlat_year_price_1: -year_2000 + year_2013 >= 0.1
ArtificialNotFlat_year_price_2: -year_2013 + year_2022 >= 0.1
ArtificialNotDictatorial_price_mileage_km_1: -mileage_km_13000 + mileage_km_374000 + price_1700 - price_39850 <=
0.95
ArtificialNotDictatorial_price_mileage_km_2: mileage_km_13000 - mileage_km_374000 - price_1700 + price_39850 <=
0.95
ArtificialNotFlat_price_mileage_km_1: price_22500 - price_39850 >= 0.1
ArtificialNotFlat_price_mileage_km_2: price_1700 - price_22500 >= 0.1
ArtificialNotFlat_mileage_km_price_1: mileage_km_110000 - mileage_km_374000 >= 0.1
ArtificialNotFlat_mileage_km_price_2: -mileage_km_110000 + mileage_km_13000 >= 0.1
ArtificialNotDictatorial_price_engine_size_cm3_1: -engine_size_cm3_1995 + engine_size_cm3_875 + price_1700 -
price_39850 <= 0.95
ArtificialNotDictatorial_price_engine_size_cm3_2: engine_size_cm3_1995 - engine_size_cm3_875 - price_1700 +
price_39850 <= 0.95
ArtificialNotFlat_price_engine_size_cm3_1: price_22500 - price_39850 >= 0.1
ArtificialNotFlat_price_engine_size_cm3_2: price_1700 - price_22500 >= 0.1
ArtificialNotFlat_engine_size_cm3_price_1: engine_size_cm3_1198 - engine_size_cm3_875 >= 0.1
ArtificialNotFlat_engine_size_cm3_price_2: -engine_size_cm3_1198 + engine_size_cm3_1995 >= 0.1
ArtificialNotDictatorial_price_power_hp_1: -power_hp_200 + power_hp_54 + price_1700 - price_39850 <= 0.95
ArtificialNotDictatorial_price_power_hp_2: power_hp_200 - power_hp_54 - price_1700 + price_39850 <= 0.95
ArtificialNotFlat_price_power_hp_1: price_22500 - price_39850 >= 0.1
ArtificialNotFlat_price_power_hp_2: price_1700 - price_22500 >= 0.1
ArtificialNotFlat_power_hp_price_1: -power_hp_54 + power_hp_85 >= 0.1
ArtificialNotFlat_power_hp_price_2: power_hp_200 - power_hp_85 >= 0.1
ArtificialNotDictatorial_year_mileage_km_1: -mileage_km_13000 + mileage_km_374000 - year_2000 + year_2022 <= 0.95
ArtificialNotDictatorial_year_mileage_km_2: mileage_km_13000 - mileage_km_374000 + year_2000 - year_2022 <= 0.95
ArtificialNotFlat_year_mileage_km_1: -year_2000 + year_2013 >= 0.1
ArtificialNotFlat_year_mileage_km_2: -year_2013 + year_2022 >= 0.1
ArtificialNotFlat_mileage_km_year_1: mileage_km_110000 - mileage_km_374000 >= 0.1
ArtificialNotFlat_mileage_km_year_2: -mileage_km_110000 + mileage_km_13000 >= 0.1
ArtificialNotDictatorial_year_engine_size_cm3_1: -engine_size_cm3_1995 + engine_size_cm3_875 - year_2000 +
year_2022 <= 0.95
ArtificialNotDictatorial_year_engine_size_cm3_2: engine_size_cm3_1995 - engine_size_cm3_875 + year_2000 -
year_2022 <= 0.95
ArtificialNotFlat_year_engine_size_cm3_1: -year_2000 + year_2013 >= 0.1
ArtificialNotFlat_year_engine_size_cm3_2: -year_2013 + year_2022 >= 0.1
ArtificialNotFlat_engine_size_cm3_year_1: engine_size_cm3_1198 - engine_size_cm3_875 >= 0.1

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ArtificialNotFlat_engine_size_cm3_year_2: -engine_size_cm3_1198 + engine_size_cm3_1995 >= 0.1
ArtificialNotDictatorial_year_power_hp_1: -power_hp_200 + power_hp_54 - year_2000 + year_2022 <= 0.95
ArtificialNotDictatorial_year_power_hp_2: power_hp_200 - power_hp_54 + year_2000 - year_2022 <= 0.95
ArtificialNotFlat_year_power_hp_1: -year_2000 + year_2013 >= 0.1
ArtificialNotFlat_year_power_hp_2: -year_2013 + year_2022 >= 0.1
ArtificialNotFlat_power_hp_year_1: -power_hp_54 + power_hp_85 >= 0.1
ArtificialNotFlat_power_hp_year_2: power_hp_200 - power_hp_85 >= 0.1
ArtificialNotDictatorial_mileage_km_engine_size_cm3_1: -engine_size_cm3_1995 + engine_size_cm3_875 +
mileage_km_13000 - mileage_km_374000 <= 0.95
ArtificialNotDictatorial_mileage_km_engine_size_cm3_2: engine_size_cm3_1995 - engine_size_cm3_875 -
mileage_km_13000 + mileage_km_374000 <= 0.95
ArtificialNotFlat_mileage_km_engine_size_cm3_1: mileage_km_110000 - mileage_km_374000 >= 0.1
ArtificialNotFlat_mileage_km_engine_size_cm3_2: -mileage_km_110000 + mileage_km_13000 >= 0.1
ArtificialNotFlat_engine_size_cm3_mileage_km_1: engine_size_cm3_1198 - engine_size_cm3_875 >= 0.1
ArtificialNotFlat_engine_size_cm3_mileage_km_2: -engine_size_cm3_1198 + engine_size_cm3_1995 >= 0.1
ArtificialNotDictatorial_mileage_km_power_hp_1: mileage_km_13000 - mileage_km_374000 - power_hp_200 + power_hp_54
<= 0.95
ArtificialNotDictatorial_mileage_km_power_hp_2: -mileage_km_13000 + mileage_km_374000 + power_hp_200 - power_hp_54
<= 0.95
ArtificialNotFlat_mileage_km_power_hp_1: mileage_km_110000 - mileage_km_374000 >= 0.1
ArtificialNotFlat_mileage_km_power_hp_2: -mileage_km_110000 + mileage_km_13000 >= 0.1
ArtificialNotFlat_power_hp_mileage_km_1: -power_hp_54 + power_hp_85 >= 0.1
ArtificialNotFlat_power_hp_mileage_km_2: power_hp_200 - power_hp_85 >= 0.1
ArtificialNotDictatorial_engine_size_cm3_power_hp_1: engine_size_cm3_1995 - engine_size_cm3_875 - power_hp_200 +
power_hp_54 <= 0.95
ArtificialNotDictatorial_engine_size_cm3_power_hp_2: -engine_size_cm3_1995 + engine_size_cm3_875 + power_hp_200 -
power_hp_54 <= 0.95
ArtificialNotFlat_engine_size_cm3_power_hp_1: engine_size_cm3_1198 - engine_size_cm3_875 >= 0.1
ArtificialNotFlat_engine_size_cm3_power_hp_2: -engine_size_cm3_1198 + engine_size_cm3_1995 >= 0.1
ArtificialNotFlat_power_hp_engine_size_cm3_1: -power_hp_54 + power_hp_85 >= 0.1
ArtificialNotFlat_power_hp_engine_size_cm3_2: power_hp_200 - power_hp_85 >= 0.1

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All Variables and Values:

```

Alternative_0_Utility = 0.3000
Alternative_10_Utility = 0.3000
Alternative_11_Utility = 0.4000
Alternative_12_Utility = 0.3000
Alternative_13_Utility = 0.2100
Alternative_14_Utility = 0.3000
Alternative_15_Utility = 0.6000
Alternative_16_Utility = 0.4000
Alternative_17_Utility = 0.2000
Alternative_18_Utility = 0.3000
Alternative_19_Utility = 0.4000
Alternative_1_Utility = 0.4000
Alternative_20_Utility = 0.3000
Alternative_21_Utility = 0.3000
Alternative_22_Utility = 0.3000
Alternative_23_Utility = 0.3000
Alternative_24_Utility = 0.4000
Alternative_25_Utility = 0.3000
Alternative_26_Utility = 0.2000
Alternative_27_Utility = 0.3000
Alternative_28_Utility = 0.3000
Alternative_29_Utility = 0.2000
Alternative_2_Utility = 0.4000
Alternative_30_Utility = 0.4000
Alternative_31_Utility = 0.8000
Alternative_32_Utility = 0.4000
Alternative_33_Utility = 0.2000
Alternative_34_Utility = 0.3000
Alternative_35_Utility = 0.3000
Alternative_36_Utility = 0.5000
Alternative_37_Utility = 0.2000
Alternative_38_Utility = 0.2000
Alternative_39_Utility = 0.5000
Alternative_3_Utility = 0.4000
Alternative_40_Utility = 0.5000
Alternative_41_Utility = 0.3000
Alternative_42_Utility = 0.3000
Alternative_43_Utility = 0.3000
Alternative_44_Utility = 0.2000
Alternative_45_Utility = 0.3000
Alternative_46_Utility = 0.4000
Alternative_47_Utility = 0.1000
Alternative_48_Utility = 0.4000
Alternative_49_Utility = 0.3000
Alternative_4_Utility = 0.3000
Alternative_5_Utility = 0.3000
Alternative_6_Utility = 0.3000
Alternative_7_Utility = 0.4000

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```
Alternative_8_Utility = 0.3100
Alternative_9_Utility = 0.3000
engine_size_cm3_1084 = 0.0000
engine_size_cm3_1108 = 0.0000
engine_size_cm3_1124 = 0.0000
engine_size_cm3_1149 = 0.0000
engine_size_cm3_1193 = 0.1000
engine_size_cm3_1197 = 0.1000
engine_size_cm3_1198 = 0.1000
engine_size_cm3_1199 = 0.1000
engine_size_cm3_1240 = 0.1000
engine_size_cm3_1242 = 0.1000
engine_size_cm3_1248 = 0.1000
engine_size_cm3_1328 = 0.1000
engine_size_cm3_1390 = 0.1000
engine_size_cm3_1398 = 0.1000
engine_size_cm3_1595 = 0.1000
engine_size_cm3_1798 = 0.1000
engine_size_cm3_1984 = 0.1000
engine_size_cm3_1995 = 0.2000
engine_size_cm3_875 = 0.0000
engine_size_cm3_898 = 0.0000
engine_size_cm3_995 = 0.0000
engine_size_cm3_998 = 0.0000
engine_size_cm3_999 = 0.0000
mileage_km_101000 = 0.1000
mileage_km_110000 = 0.1000
mileage_km_111200 = 0.0000
mileage_km_122185 = 0.0000
mileage_km_130000 = 0.2000
mileage_km_132000 = 0.0000
mileage_km_139990 = 0.0000
mileage_km_145500 = 0.0000
mileage_km_152000 = 0.0000
mileage_km_153000 = 0.0000
mileage_km_153533 = 0.0000
mileage_km_154400 = 0.0000
mileage_km_158000 = 0.0000
mileage_km_159000 = 0.0000
mileage_km_160000 = 0.0000
mileage_km_163035 = 0.0000
mileage_km_164000 = 0.0000
mileage_km_165000 = 0.0000
mileage_km_173117 = 0.0000
mileage_km_177491 = 0.0000
mileage_km_177896 = 0.0000
mileage_km_181500 = 0.0000
mileage_km_184000 = 0.0000
mileage_km_199700 = 0.0000
mileage_km_232000 = 0.0000
mileage_km_281980 = 0.0000
mileage_km_34900 = 0.2000
mileage_km_374000 = 0.0000
mileage_km_39000 = 0.2000
mileage_km_40400 = 0.2000
mileage_km_42000 = 0.2000
mileage_km_48000 = 0.2000
mileage_km_49459 = 0.1100
mileage_km_56213 = 0.1100
mileage_km_57000 = 0.1000
mileage_km_58000 = 0.1000
mileage_km_60098 = 0.1000
mileage_km_65900 = 0.1000
mileage_km_68000 = 0.1000
mileage_km_71662 = 0.1000
mileage_km_78000 = 0.1000
mileage_km_79000 = 0.1000
mileage_km_82000 = 0.1000
mileage_km_85000 = 0.1000
mileage_km_86000 = 0.1000
mileage_km_87975 = 0.1000
mileage_km_88000 = 0.1000
mileage_km_93000 = 0.1000
mileage_km_99600 = 0.1000
power_hp_102 = 0.1000
power_hp_105 = 0.1000
power_hp_111 = 0.1000
power_hp_118 = 0.1000
power_hp_122 = 0.1000
power_hp_143 = 0.1000
power_hp_160 = 0.1000
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```
power_hp_200 = 0.2000
power_hp_54 = 0.0000
power_hp_58 = 0.0000
power_hp_60 = 0.0000
power_hp_62 = 0.0000
power_hp_65 = 0.0000
power_hp_67 = 0.0000
power_hp_68 = 0.0000
power_hp_69 = 0.0000
power_hp_75 = 0.0000
power_hp_80 = 0.1000
power_hp_82 = 0.1000
power_hp_85 = 0.1000
power_hp_90 = 0.1000
power_hp_92 = 0.1000
power_hp_95 = 0.1000
price_10500 = 0.2000
price_11800 = 0.2000
price_12999 = 0.2000
price_13900 = 0.2000
price_13999 = 0.2000
price_14600 = 0.1000
price_15000 = 0.1000
price_16500 = 0.1000
price_1700 = 0.2000
price_17700 = 0.1000
price_18700 = 0.1000
price_19000 = 0.1000
price_19300 = 0.1000
price_21300 = 0.1000
price_21900 = 0.1000
price_22500 = 0.1000
price_24900 = 0.0000
price_25900 = 0.0000
price_26000 = 0.0000
price_26900 = 0.0000
price_27700 = 0.0000
price_27999 = 0.0000
price_29800 = 0.0000
price_29900 = 0.0000
price_30500 = 0.0000
price_31000 = 0.0000
price_31900 = 0.0000
price_32700 = 0.0000
price_36900 = 0.0000
price_37900 = 0.0000
price_38000 = 0.0000
price_38500 = 0.0000
price_39360 = 0.0000
price_39500 = 0.0000
price_39850 = 0.0000
price_4700 = 0.2000
price_4900 = 0.2000
price_7400 = 0.2000
price_7800 = 0.2000
price_9000 = 0.2000
year_2000 = 0.0000
year_2001 = 0.0000
year_2004 = 0.0000
year_2006 = 0.0000
year_2007 = 0.0000
year_2009 = 0.0000
year_2010 = 0.0000
year_2011 = 0.0000
year_2012 = 0.0000
year_2013 = 0.1000
year_2014 = 0.1000
year_2015 = 0.1000
year_2016 = 0.1000
year_2017 = 0.1000
year_2018 = 0.2000
year_2019 = 0.2000
year_2021 = 0.2000
year_2022 = 0.2000
```

Objective Value: 1.5000

Criterion Weights (Maximum Utility Value):

Weight of price: 0.2000

Weight of year: 0.2000

Weight of mileage_km: 0.2000

Weight of engine_size_cm3: 0.2000
Weight of power_hp: 0.2000

Alternative Rankings:

Rank	Car	Utility	Alternative
0	1 Mitsubishi Space Star 1.2 Clear Tec CVT Active+	0.80	Alternative_31
1	2 Peugeot 208 PureTech 75 Allure	0.60	Alternative_15
2	3 BMW Seria 1 118i	0.50	Alternative_40
3	4 Citroën C3 1.2 VTi Attraction	0.50	Alternative_36
4	5 Fiat 500 1.2 8V Start&Stopp Collezione	0.50	Alternative_39
5	6 Peugeot 208 1.2 PureTech Style	0.40	Alternative_30
6	7 Audi A1 1.4 TFSI Sportback S tronic Ambition	0.40	Alternative_16
7	8 Volkswagen Polo 1.2 TSI BMT Highline	0.40	Alternative_32
8	9 Suzuki Swift 1.0 T Elegance	0.40	Alternative_11
9	10 Kia Rio	0.40	Alternative_24
10	11 Renault Clio (Energy) TCe 90 Start & Stop LIMITED	0.40	Alternative_19
11	12 Fiat 500 1.2 8V Anniversario	0.40	Alternative_7
12	13 Renault Megane 1.4 RN 16V	0.40	Alternative_1
13	14 Peugeot 208 1.2 PureTech Active	0.40	Alternative_46
14	15 Nissan Micra 1.2 City	0.40	Alternative_48
15	16 Mitsubishi Space Star	0.40	Alternative_3
16	17 Dacia Sandero Stepway TCe 90 (S&S) Essential	0.40	Alternative_2
17	18 Kia Picanto	0.31	Alternative_8
18	19 Opel Corsa 1.4 Cosmo	0.30	Alternative_34
19	20 Renault Clio 0.9 Energy TCe Limited	0.30	Alternative_0
20	21 Skoda Fabia 1.0 Active	0.30	Alternative_35
21	22 Opel Corsa	0.30	Alternative_27
22	23 Volkswagen up! 1.0 Black Style	0.30	Alternative_41
23	24 Renault Twingo	0.30	Alternative_42
24	25 Audi A3 2.0 TFSI Quattro Ambition S tronic	0.30	Alternative_43
25	26 Suzuki Swift 1.3 Comfort	0.30	Alternative_45
26	27 Peugeot 206 plus 206+ 60 Generation	0.30	Alternative_28
27	28 Volkswagen Polo 1.2 TSI Blue Motion Technology...	0.30	Alternative_25
28	29 Audi A3	0.30	Alternative_14
29	30 Volkswagen Golf	0.30	Alternative_12
30	31 Nissan Micra	0.30	Alternative_4
31	32 Ford KA	0.30	Alternative_5
32	33 Alfa Romeo Mito 0.9 TwinAir Progression	0.30	Alternative_6
33	34 Audi A3	0.30	Alternative_9
34	35 Renault Clio (Energy) TCe 90 Bose Edition	0.30	Alternative_10
35	36 Ford Fiesta	0.30	Alternative_49
36	37 Fiat 500 C 0.9 8V TwinAir Start&Stopp	0.30	Alternative_18
37	38 Kia Picanto	0.30	Alternative_20
38	39 Renault Twingo	0.30	Alternative_21
39	40 Chevrolet Spark 1.0 Base	0.30	Alternative_22
40	41 Renault Captur 1.2 Energy TCe Intens EDC EU6	0.30	Alternative_23
41	42 Toyota Yaris 1.0 Active	0.21	Alternative_13
42	43 Volkswagen up! (BlueMotion Technology) move	0.20	Alternative_17
43	44 Renault Twingo	0.20	Alternative_37
44	45 Ford Fiesta 1.1 Connected	0.20	Alternative_38
45	46 Citroën C1 1.0 VTi Shine EU6	0.20	Alternative_29
46	47 Toyota Yaris 1.0 Luna A/C	0.20	Alternative_33
47	48 Fiat Panda 1.1 Active Plus	0.20	Alternative_44
48	49 Mitsubishi Colt 1.1 ClearTec Edition	0.20	Alternative_26
49	50 Toyota Yaris	0.10	Alternative_47

Appendix B

AHP Output

	Reliability	Cost	Performance
Reliability	1.000000	3.000000	5.0
Cost	0.333333	1.000000	3.0
Performance	0.250000	0.333333	1.0
category weights: ['0.6284', '0.2578', '0.1138']			
consistency ratio: 0.1174 (inconsistent!)			
reconstructed			
	Reliability	Cost	Performance
Reliability	1.000000	2.437196	5.523185
Cost	0.410308	1.000000	2.266205
Performance	0.181055	0.441266	1.000000
difference			
	Reliability	Cost	Performance
Reliability	0.000000	0.562804	-0.523185

Cost	-0.076974	0.000000	0.733795
Performance	0.068945	-0.107933	0.000000
largest difference: 0.7338 at position Cost vs Performance			

performance criteria			
	Engine Size	Power	
Engine Size	1.0	0.333333	
Power	3.0	1.000000	
performance criteria weights: ['0.2500', '0.7500']			
consistency ratio: 0.0000 (consistent)			

reliability criteria			
	Mileage	Year	
Mileage	1.000000	3.0	
Year	0.333333	1.0	
reliability criteria weights: ['0.7500', '0.2500']			
consistency ratio: 0.0000 (consistent)			

Price: 0.2578			
Engine Size: 0.0284			
Power: 0.0853			
Mileage: 0.4713			
Year: 0.1571			
alternative Evaluation			

	Rank	Car	Score
0	1	Mitsubishi Space Star 1.2 Clear Tec CVT Active+	0.786072
1	2	Kia Picanto	0.782069
2	3	Peugeot 208 PureTech 75 Allure	0.777165
3	4	Nissan Micra 1.2 City	0.764807
4	5	Peugeot 206 plus 206+ 60 Generation	0.760613
5	6	Chevrolet Spark 1.0 Base	0.744914
6	7	BMW Seria 1 118i	0.729143
7	8	Fiat 500 1.2 8V Anniversario	0.725468
8	9	Kia Rio	0.702719
9	10	Renault Megane 1.4 RN 16V	0.696382
10	11	Volkswagen up! 1.0 Black Style	0.695548
11	12	Fiat 500 1.2 8V Start&Stopp Collezione	0.686680
12	13	Renault Twingo	0.685740
13	14	Renault Twingo	0.676993
14	15	Peugeot 208 1.2 PureTech Style	0.672051
15	16	Toyota Yaris 1.0 Active	0.666373
16	17	Mitsubishi Space Star	0.665424
17	18	Suzuki Swift 1.3 Comfort	0.662999
18	19	Volkswagen Polo 1.2 TSI BMT Highline	0.661662
19	20	Fiat Panda 1.1 Active Plus	0.661169
20	21	Renault Clio (Energy) TCe 90 Start & Stop LIMITED	0.658970
21	22	Toyota Yaris 1.0 Luna A/C	0.647252
22	23	Citroën C3 1.2 VTi Attraction	0.643477
23	24	Toyota Yaris	0.642423
24	25	Renault Twingo	0.639841
25	26	Renault Clio (Energy) TCe 90 Bose Edition	0.635085
26	27	Mitsubishi Colt 1.1 ClearTec Edition	0.632874
27	28	Ford KA	0.631470
28	29	Fiat 500 C 0.9 8V TwinAir Start&Stopp	0.630229
29	30	Kia Picanto	0.619146
30	31	Audi A3 2.0 TFSI Quattro Ambition S tronic	0.618279
31	32	Citroën C1 1.0 VTi Shine EU6	0.615726
32	33	Nissan Micra	0.606163
33	34	Renault Clio 0.9 Energy TCe Limited	0.605738
34	35	Ford Fiesta	0.605508
35	36	Volkswagen up! (BlueMotion Technology) move	0.604093
36	37	Audi A1 1.4 TFSI Sportback S tronic Ambition	0.603806
37	38	Dacia Sandero Stepway TCe 90 (S&S) Essential	0.599488
38	39	Alfa Romeo Mito 0.9 TwinAir Progression	0.591136
39	40	Volkswagen Golf	0.585249
40	41	Suzuki Swift 1.0 T Elegance	0.574364
41	42	Peugeot 208 1.2 PureTech Active	0.567927
42	43	Skoda Fabia 1.0 Active	0.565875
43	44	Opel Corsa 1.4 Cosmo	0.563319
44	45	Volkswagen Polo 1.2 TSI Blue Motion Technology...	0.548585
45	46	Ford Fiesta 1.1 Connected	0.546750
46	47	Opel Corsa	0.542788
47	48	Renault Captur 1.2 Energy TCe Intens EDC EU6	0.541444
48	49	Audi A3	0.501520

```
49      50      Audi A3  0.361544  
visualizations saved as 'ahp_weights.png' and 'consistency_ratios.png'
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